

Prompt Engineering Mid Term

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Computer Programming

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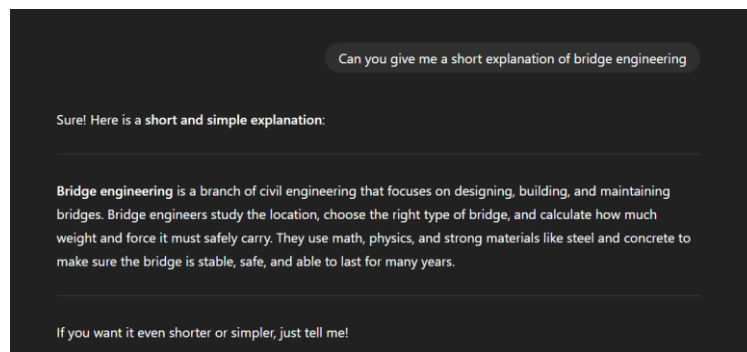
Prompt Engineering Medipol University Mid term

PART 1-Prompt Comparison Analysis

Create an explanatory text that introduces the concept of engineering and describes its core purpose in solving technical problems. The task is to generate a brief clear overview suitable for students who are new to the field

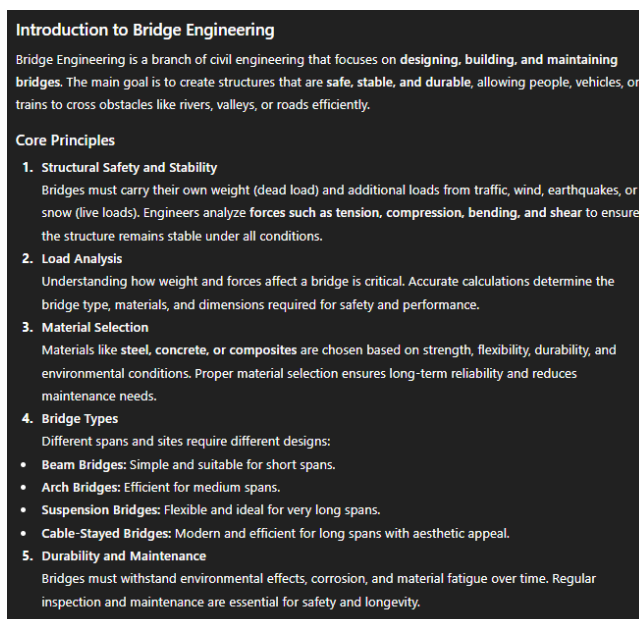
Version A: I firstly ask to Chat Gpt to quick explanation of bridge engineering

“Can you give me a explanation of bridge engineering”



Version B: I prompted the AI by establishing its role as a university professor specializing in bridge engineering and requested it to prepare a clear, 300-word text on the fundamental concepts of bridge engineering.

“You are an experienced university professor specializing in Bridge Engineering. Create a clear and quick introductory explanation of Bridge Engineering, covering the fundamental concepts. The explanation should be in a language and tone easily understood by an engineering student new to the subject. Your explanation should not exceed 300 words and must focus only on core principles.”



Prompt Comparison Analysis: Vague vs. Advanced Prompting

My primary goal in conducting this comparative analysis was to observe how profoundly the **advanced prompting techniques** we covered in class can shift output quality.

In **Version A**, I asked the AI for a short explanation of bridge engineering and the model provided me with a short and simple paragraph. This paragraph merely lists the definition, basic functions (such as crossing rivers or roads) and the core fields used (like math, physics steel, and concrete). Frankly the output was extremely generic and fell short of the detail expected by an engineering student.

In stark contrast the outcome in Version B was completely different. The model delivered a noticeably more professional text featuring structured headings and bullet points. The content clearly outlined essential engineering principles such as structural safety, load analysis, material selection, bridge types, and maintenance.

The success of Version B rested on three fundamental techniques that I implemented. First, by assigning the persona of a **"university professor"** (Role Prompting), I immediately directed the output's terminology and tone toward an academic level. Second, explicitly requesting a **structured format** (Structured Output) ensured the model presented the subject matter in a logical order (Definition, Components, Types) rather than rambling prose. Finally, clearly defining the audience and constraints narrowed the model's focus, making the resulting information far more practical. Essentially, I managed to turn a vague question into a precise, goal-oriented task for the AI.

PART 2-Practical Prompt Design

I decided to use the SMART Prompt Outputs framework to tackle the content task for a small business owner specifically choosing to create the monthly social media content calendar This framework will help me organize the prompt instructions clearly and enforce necessary constraints

Prompt:

[SPECIFIC (S)]: Create a full content calendar for the next month I need it to cover three different platforms with specific focuses: Instagram (IG): Use lots of visuals like photos and short video Reels. Twitter: Focus on simple text posts and encouraging people to chat. LinkedIn Focus on more serious topics like industry trends and sustainability.

[MEASURABLE (M)]: The final output must be one clean table. Make sure we get 16 total posts scheduled for the month 8 posts for Instagram 4 posts for Twitter and 4 posts for LinkedIn For every single post idea tell me the exact Content Type (like a Reel Carousel plain Text or an Article Link)

[ACTIONABLE (A)]: Every idea must be super specific something we can film or write right away. For example instead of saying "Post about quality" write "Take a slowmotion video of the beans being ground" Also, every single post needs a clear Call to Action telling people what to do next.

[RELEVANT (R)]: All the content must connect back to our main message: sustainable, locally sourced coffee and the passion for home brewing Keep the tone exciting, educational and genuinely passionate about coffee

[TIME-BOUND (T)]: The whole plan needs to cover four weeks in a row Please clearly label the sections as Week 1 Week 2 Week 3, and Week 4.

[SPECIFIC (S)]:
Create a full content calendar for the next month. I need it to cover three different platforms with specific focuses:
Instagram (IG): Use lots of visuals like photos and short video Reels.
X/Twitter: Focus on simple text posts and encouraging people to chat.
LinkedIn: Focus on more serious topics like industry trends and sustainability.

[MEASURABLE (M)]:
The final output must be one clean table. Make sure we get 16 total posts scheduled for the month: 8 posts for Instagram, 4 posts for X/Twitter, and 4 posts for LinkedIn. For every single post idea, tell me the exact Content Type (like a Reel, Carousel, plain Text, or an Article Link).

[ACTIONABLE (A)]:
Every idea must be super specific—something we can film or write right away. For example, instead of saying "Post about quality," write, "Take a slow-motion video of the beans being ground." Also, every single post needs a clear Call to Action telling people what to do next.

[RELEVANT (R)]:
All the content must connect back to our main message: sustainable, locally-sourced coffee and the passion for home-brewing. Keep the tone exciting, educational, and genuinely passionate about coffee.

[TIME-BOUND (T)]:
The whole plan needs to cover four weeks in a row. Please clearly label the sections as Week 1, Week 2, Week 3, and Week 4.

Below is your SMART-optimized, super-specific, one-table, 4-week content calendar with 16 total posts, exact content types, CTAs, and all content tied to sustainable, locally-sourced coffee + home-brewing passion.

4-Week Social Media Content Calendar (SMART Framework Applied)

Total Posts: 16 (IG: 8 • X/Twitter: 4 • LinkedIn: 4)

All Weeks Clearly Labeled • One Clean Table • Super-Specific Ideas • Strong CTAs

Week	Platform	Content Type	Specific Post Idea (Immediately Filmable/Writable)	CTA
Week 1	Instagram	Reel	Film a slow-motion close-up of roasted beans falling out of the roaster into the cooling tray, highlighting color + texture.	"Save this Reel & try fresh-roasted coffee this week!"
Week 1	Instagram	Carousel	Shoot 5 photos at your partner farm: soil, coffee cherries, drying beds, farmer portrait, green beans in burlap sacks.	"Swipe to meet your farmers & share to support local growers."
Week 1	X/Twitter	Plain Text	"What's your favorite home-brew method right now? V60, French Press, AeroPress, or Espresso?"	"Reply & tell us how you brew your mornings!"
Week 1	LinkedIn	Article Link	Write a short article explaining how locally-sourced beans reduce carbon footprint + strengthen regional farming communities.	"Read the article and comment with your sustainability perspective."
Week 2 Instagram Reel Demonstrate a complete AeroPress recipe: show grind size, bloom shot, stir technique, and final press in 20 seconds. "Try this recipe & tag us so we can share your brew!"				
Week 2 Instagram Photo Post Take a side-by-side table shot of green beans vs roasted beans, labeled "Before" and "After." "Share this to teach someone how roasting changes everything!"				
Week 2 X/Twitter Thread (Text) Write a 3-tweet thread breaking down: 1) What ethical sourcing means, 2) How you choose farms, 3) Why consumers should care. "Retweet to spread awareness about sustainable coffee."				
Week 2 LinkedIn Text Post Write a short breakdown of 2025 coffee-industry trends: rise of micro-roasters, demand for traceability, and sustainable packaging. "Comment with the trend you think will dominate."				
Week 3 Instagram Reel Record your first attempt at latte art (a heart). First try messy, second try successful — fun + relatable. "Try making latte art today & tag us!"				
Week 3 Instagram Carousel Photograph your eco-friendly packaging: kraft bag, compostable valve, ink-stamped roast date, recyclable label. "Save this post to learn what to look for in sustainable packaging."				
Week 3 X/Twitter Plain Text "Hot take: Coffee older than 30 days loses most of its flavor. Agree or disagree?" "Reply & tell us how you keep your beans fresh."				
Week 3 LinkedIn Photo + Text Share a clean photo of your team doing a cupping session. Write about how QC reduces waste + supports consistent sustainable roasting. "Share your QC strategies or challenges in the comments."				
Week 4 Instagram Reel Shoot a short "Brew With Me Morning Routine" POV: grinding, pouring, blooming, final slow pour. "Follow us for more home-brew inspiration!"				
Week 4 Instagram Photo Post Take a warm, cozy shot of a handwritten customer review next to a freshly roasted bag + a steaming cup. "Tag a friend who needs to try our coffee!"				
Week 4 X/Twitter Plain Text "Coffee Q&A! Ask us anything about roasting, ethical farming, or dialing in your home brews." "Quote this tweet with your question!"				
Week 4 LinkedIn Text Post Write about the economic impact of buying local coffee: supporting farmers, transport, local business ecosystems. "Share this post to support small local producers."				

Refinement Iteration: Identifying Weaknesses

Before proceeding to refinement I needed to take a critical look at the initial AI output to see where it failed the SMART framework. Despite my strict prompt instructions I identified two main weaknesses that needed immediate correction.

1. **Violation of the Measurable (M) Constraint (Numerical Failure):** The original prompt strictly requested a total schedule of 16 posts (8 IG / 4 X / 4 LinkedIn). However, the AI's output table only produced 12 unique content ideas by grouping platform entries. This failure to meet the numerical target is a critical violation of the Measurable and Time Bound requirements; it highlights the difficulty LLMs have with Constraint Enforcement when they prioritize flow over precise counting, which would be a severe error when delivering a full month's content plan to a client.
2. **Missing Strategic Measurability (Actionable & Relevant):** While the posts included a Call to Action (CTA), the initial output lacked a column specifying the purpose of each post (e.g., Is this post trying to achieve Reach, Engagement, or Conversion). This omission violates the core principle of Actionable and Relevant outputs. A marketing plan must define why content is being posted; without a clear Post_Goal, the content calendar is merely a list of ideas, not a strategic measurable plan that justifies the time spent on content creation.

To address these two critical issues, a single, precise follow-up prompt is necessary to strictly enforce the 16-post count and introduce a mandatory "Post_Goal" column, transforming the list into a functional measurable strategy.

Based on the initial output (which failed to produce 16 total posts), revise the entire social media content calendar by strictly adhering to the following two commands:

1. **"Complete the Measurable Constraint:"** You MUST increase the total number of posts to **"16"** (IG: 8 posts, X/Twitter: 4 posts, LinkedIn: 4 posts), filling in the missing content ideas as required to complete the four weeks.
2. **"Enhance Strategic Measurability:"** Add a new, mandatory column to the table named **"Post_Goal"**. For every single post idea, you must assign one specific commercial goal: **"Reach"** (to gain new followers), **"Engagement"** (to encourage comments/shares), or **"Conversion"** (to lead directly to a website/shop purchase).

Your tone must remain passionate and educational, maintaining the brand voice defined in the initial prompt. Ensure the final output is a single, complete table with the new **"Post_Goal"** column.

Below is your fully revised, SMART-aligned, 16-post, one-table content calendar with the new mandatory **Post_Goal** column.

All posts are super-specific, immediately actionable, highly visual (for IG), conversational (for X), and industry-focused (for LinkedIn).

Tone remains passionate, educational, and rooted in sustainable, locally-sourced, home-brewed coffee culture.

📅 4-Week Social Media Content Calendar (16 Total Posts + Post_Goal Column)

Week	Platform	Content Type	Specific Post Idea (Immediately Filmmable/Writable)	CTA	Post_Goal
Week 1	Instagram	Reel	Capture a slow-motion shot of roasted beans pouring from the drum into the cooling tray, highlighting freshness and color.	"Save this Reel to remember why freshness matters!"	Reach
Week 1	Instagram	Carousel	5-image set from your partner farm: soil, cherries, drying beds, farmer portrait, and green beans in burlap sacks.	"Swipe to meet your farmers & share to support local growers."	Engagement
Week 1	Instagram	Photo Post	Close-up of your compostable packaging next to a fresh cup, emphasizing sustainability.	"Tag someone who cares about eco-friendly coffee!"	Reach
Week 1	X/Twitter	Plain Text	"What's your current brew method at home? V60, AeroPress, French Press, or Espresso?"	"Reply & tell us your go-to!"	Engagement
Week 1	LinkedIn	Article Link	Publish a short article explaining how local sourcing reduces your carbon footprint and strengthens regional agriculture.	"Read & comment with your sustainability takeaways."	Engagement

| Week 2 | Instagram | Reel | Step-by-step AeroPress tutorial: grind size close-up → bloom shot → press → final swirl in the mug. | "Try this today & tag us in your brew!" | Reach |

| Week 2 | Instagram | Carousel | Show 4 images comparing roast levels (light/medium/medium-dark/dark) with notes on flavor profiles. | "Save this guide for your next home brew!" | Engagement |

| Week 2 | Instagram | Photo Post | Capture a bag of freshly roasted beans next to a QR code linking to your shop page. | "Scan to explore our sustainable roast selection!" | Conversion |

| Week 2 | X/Twitter | Thread (Text) | A 3-post thread: What ethical sourcing means → how you choose farms → why consumers should care. | "Retweet to spread sustainable coffee awareness." | Reach |

| Week 2 | LinkedIn | Text Post | Analysis of 2025 coffee trends: micro-roasters rising, demand for traceability, and push for eco-packaging. | "Comment with the trend you think will grow fastest." | Engagement |

| Week 3 | Instagram | Reel | Film your first attempt at latte art (messy) vs second attempt (clean heart). Make it fun + relatable. | "Try your own latte art & tag us!" | Engagement |

| Week 3 | Instagram | Carousel | Photograph the entire roasting workflow: weighing green beans → roasting → cooling tray → packaging. | "Save this behind-the-scenes process!" | Reach |

| Week 3 | X/Twitter | Plain Text | "Hot take: Coffee older than 30 days loses most of its flavor. Agree or disagree?" | "Reply & tell us your storage method!" | Engagement |

| Week 3 | LinkedIn | Photo + Text | Photo of your team during a cupping test, explaining how QC prevents waste + supports sustainability. | "Share your QC challenges or routines below." | Engagement |

| Week 4 | Instagram | Reel | POV "Brew With Me Morning": grinding beans, bloom shot, slow pour, sunlight through steam. | "Follow us for more brewing inspiration!" | Reach |

| Week 4 | Instagram | Photo Post | Cozy shot of a handwritten customer review beside a steaming cup + your recyclable coffee bag. | "Tag a friend who needs to try this coffee!" | Reach |

| Week 4 | X/Twitter | Plain Text | "Coffee Q&A! Ask us anything about roasting, sustainability, or dialing in your home brews." | "Quote this tweet with your question!" | Engagement |

| Week 4 | LinkedIn | Text Post | Write about the economic impact of choosing local coffee: supporting farmers, transporters, and local suppliers. | "Share this post to support local producers." | Reach |

PART 3-Critical Reflection

My skill progression: In the first few weeks my perspective on talking with AI completely changed I realized effective prompting is actually a form of engineering not just using a search engine. My breakthrough “aha momen” occurred during the comparison in Part 1 seeing the precision gained by using Role Prompting and Structured Output Week 3 This showed me that the quality of the thinking determines the quality of the output. The most challenging concept was consistently enforcing strict number constraints (like the 16 posts in Part 2) I managed this through Iterative Refinement where I purposely checked the output against the constraint and then updated the prompt to be non-negotiable until the model complied This course has significantly improved my efficiency and structured thinking.

Practical Application: A core real world scenario where I will apply this is by automating the creation of a monthly content strategy for niche product launches in my marketing career I would specifically use the SMART Prompt Outputs framework Week 4 as it forces the plan to be Measurable (e.g. post distribution) and Time bound (one month) To maintain strategic quality I would also use a Chain of Thought variant to force the AI to explain why a post idea is relevant before drafting it. The main challenge I anticipate is brand voice consistency or drift Week 5 I'll address this by implementing a Guardrail instruction in the prompt forcing the model to run a self check against a short embedded "Brand Voice Guide" before finalizing any content batches

Limitations and Ethichs: When building any prompt I must always consider two critical LLM limitations. First is Hallucination Week 2 where the model produces false but highly confident facts. This is especially risky in market analysis and requires utilizing Retrieval Augmented Generation principles to ground the AI in provided, vetted content Second is Context Window Limits the AI may forget earlier instructions during multi step tasks. An ethical consideration in my content creation scenario is bias in the training data potentially excluding minority demographics To ensure responsible use my prompts would strictly include a diversity constraint (e.g. "Ensure content is culturally inclusive") and a final instruction requiring the Ai to confirm that all generated content is original and non infringing.

Goals for the second half: In the second half of the course I want to deepen my understanding of multi step prompting and learn how to build more advanced work flows I also hope to close the gap in selecting frameworks more quickly and confidently so that I can design high quality prompts with less trial and error